

Solutions to MP204 Problem Set 3

①

P.1

To do this problem, we'll need the electrical conductivities of water, platinum and gold; I found these on <http://www.earthlog.org>, a URL with lots of info about everything. The values it gives are

$$\sigma_{\text{H}_2\text{O}} = 33.333 \text{ m}\Omega^{-1}\text{cm}^{-1} = 33.333 (10^{-3}\Omega)^{-1}(10^{-2}\text{m})^{-1} \\ = 3.33 \times 10^6 \Omega^{-1}\text{m}^{-1}$$

$$\sigma_{\text{Pt}} = 94.34 \text{ m}\Omega^{-1}\text{cm}^{-1} = 9.434 \times 10^6 \Omega^{-1}\text{m}^{-1}$$

$$\sigma_{\text{Au}} = 446.4 \text{ m}\Omega^{-1}\text{cm}^{-1} = 4.464 \times 10^7 \Omega^{-1}\text{m}^{-1}$$

With these, and the formula $R = \frac{L}{\sigma A}$, we can find the resistances of the given objects.

(a) $L = 1\text{cm}$ and $A = 1\text{cm}^2$, so

$$R_{\text{H}_2\text{O}} = \frac{10^{-2}\text{m}}{(3.33 \times 10^6 \Omega^{-1}\text{m}^{-1})(10^{-2}\text{m})^2} = \boxed{3 \times 10^{-5} \Omega = 30 \mu\Omega}$$

(b) $L = 1\text{m}$ and $R = 0.5\text{cm}$, so

$$R_{\text{Pt}} = \frac{1\text{m}}{(9.434 \times 10^6 \Omega^{-1}\text{m}^{-1})(\pi(0.5 \times 10^{-2}\text{m})^2)} = \boxed{1.35 \times 10^{-3} \Omega = 1.35 \text{ m}\Omega}$$

(c) $L = 10\text{cm}$, $t = 3\text{cm}$, $w = 5\text{cm}$, so

$$R_{\text{Au}} = \frac{10^{-1}\text{m}}{(4.464 \times 10^7 \Omega^{-1}\text{m}^{-1})(3 \times 10^{-2}\text{m})(5 \times 10^{-2}\text{m})} = \boxed{1.49 \times 10^{-6} \Omega = 1.49 \mu\Omega}$$

All these are quite small, showing that metals aren't very good if you want high resistances. However, they are very good if you want to build wires out of them; notice that the gold object, even though it's a pretty big chunk, still has a resistance of less than two microhms of ohm.

P.2

(a) We did a similar calculation to this in lecture, but this is slightly more general: the atomic mass number A of an element is the average number of nucleons (protons and neutrons) in a nucleus. Since almost all elements have isotopes with slightly different numbers of neutrons (e.g., ^{12}C and ^{14}C), A is often not an integer, even though it's generally close to one. In any case, if we use the asymptotic fit $m_n \approx m_p$ (which is true), then the mass of an atom of atomic mass number

(2)

α is about amp .

Now, consider a volume V of this stuff; if ρ is its mass density, then $M = \rho V$ is the total mass of this sample. How many atoms does it contain? Since each atom has mass amp , then it takes N total atoms, $M = N \text{amp}$, so $N \text{amp} = \rho V$. Thus, the number of atoms per unit volume is $N/V = \rho/\text{amp}$.

Now, suppose each atom can contribute k electrons to a current flowing through this sample. Thus, Nk gives the total number of free electrons, so $n = Nk/V$ is the conduction electron number density. But we have N/V from above, so we see

$$n = \frac{k\rho}{\text{amp}}$$

(b) We know $\vec{J} = qn\vec{v}$ for charges q moving at velocity $\vec{v}$ with number density n . Thus, if the charges are electrons, and the sample is that given in (a), then

$$\vec{J} = (-e) \left(\frac{k\rho}{\text{amp}} \right) (\vec{v}) = - \frac{k e \rho}{\text{amp}} \vec{v}$$

The magnitude of $\vec{J}$ is thus $|\vec{J}| = \frac{k e \rho}{\text{amp}} |\vec{v}|$, giving a drift speed of $|\vec{v}| = \frac{\text{amp} |\vec{J}|}{k e \rho}$. But if $|\vec{J}|$ is due to a constant current I moving through a cross-sectional area A , $|\vec{J}| = I/A$, giving

$$|\vec{v}| = \frac{\text{amp} I}{k e \rho A}$$

as a fairly handy formula relating the drift speed to the material and the current flowing through it. For indium, $\alpha = 114.818$ and $\rho = 7.31 \text{ g cm}^{-3}$ (both from Wikipedia). You're told $k = 3$, and $e = 1.6 \times 10^{-19} \text{ C}$ and $\text{amp} = 1.67 \times 10^{-27} \text{ kg}$ are fundamental constants of nature. Using these, and the given value of $I = 150 \text{ mA}$, we find for $A = 3 \text{ cm}^2$,

$$|\vec{v}| = \frac{(114.818)(1.67 \times 10^{-27} \text{ kg})(150 \times 10^{-3} \text{ A})}{(3)(1.6 \times 10^{-19} \text{ C})(7.31 \times 10^3 \text{ kg m}^{-3})(3 \times 10^{-4} \text{ m}^2)}$$

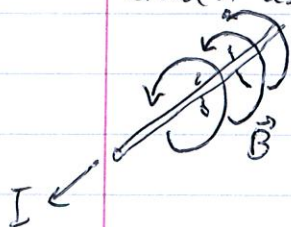
$$= 2.73 \times 10^{-8} \text{ m s}^{-1}$$

which isn't very fast at all. In fact, if this sample were 10 cm long, it would take $3.66 \times 10^6 \text{ s}$ to go from one end to the other, which is over 42 days!

(3)

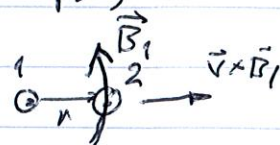
P.3

The key to taking us using the result we derived in class: if a current I flows through an infinitely long wire, the magnetic field $\vec{B}$ due to this current has magnitude $\frac{\mu_0 I}{2\pi r}$ at a perpendicular distance r from the wire, and wraps around it as shown below. We'll need to use both of these facts to



find the answers.

(a) If we take the current as coming out of the page, then the magnetic field at wire 2 due to wire 1 is indicated as shown here:



Now, if the current is 2 flows out of this page. Then these electrons are moving into the page; this gives the direction of their velocity $\vec{v}$. $\vec{v} \times \vec{B}_1$ points away from 1 as shown, but $q = -e$ for electrons, so the force of wire 1 on wire 2 is $-e \vec{v} \times \vec{B}_1$, which points toward 1. Thus, wire 2 is pulled toward wire 1 and so the two wires attract one another.

(b) For this, $|\vec{B}_1|$ is what we're after. We know $I = 1\text{ A}$ is the current flowing through both wires, and wire 2 is 1 m from wire 1, so the magnitude of the field due to wire 1 at wire 2 is

$$|\vec{B}_1| = \frac{\mu_0 I}{2\pi r} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{A}^{-1} \cdot \text{m})(1\text{ A})}{2\pi (1\text{ m})} = \boxed{2 \times 10^{-7} \text{ T}}$$

(c) Take a length L of wire 2. If it has cross-sectional area A , then the total volume of this length is AL , and the total number of electrons is $N = nAL$, where n is the electron density. But $n = \frac{|\vec{J}|}{|q||\vec{v}|}$, so $N = \frac{|\vec{J}|AL}{|q||\vec{v}|}$. $|q| = e$ for electrons, and $|\vec{J}|A = I$, the total current, so $N = \frac{IL}{e|\vec{v}|}$, giving

$$|\vec{v}| = \frac{IL}{eN}$$

as the drift speed.

(d) The force wire 2 feels due to wire 1 is, as mentioned in (a),

$$\vec{F} = -e \vec{v} \times \vec{B}_1. \text{ The magnitude is therefore } |\vec{F}| = e |\vec{v} \times \vec{B}_1|.$$

Now, $|\vec{v} \times \vec{B}_1| = |\vec{v}| |\vec{B}_1| \sin \theta$, where θ is the angle between $\vec{v}$ and $\vec{B}_1$. Hence, $\vec{v}$ is into the page and $\vec{B}_1$ points to the right, so they're at right angles to each other, i.e. $\theta = \pi/2$. Thus, $|\vec{v} \times \vec{B}_1| = |\vec{v}| |\vec{B}_1|$.

(4)

We know $|\vec{v}| = \frac{IL}{eN}$ and $|\vec{B}_1| = \frac{\mu_0 I}{2\pi r}$, so further

$$|\vec{F}| = e \left(\frac{IL}{eN} \right) \left(\frac{\mu_0 I}{2\pi r} \right) = \frac{\mu_0 I^2 L}{2\pi r N}$$

as the force that each electron feels, this was, that the total force felt by all N electrons is the length L of wire is

$$N|\vec{F}| = \frac{\mu_0 I^2 L}{2\pi r}$$

Putting in $I = 1A$, $L = 1m$, $r = 1m$ and $\mu_0 = 4\pi \times 10^{-7} T \cdot m \cdot A^{-1}$

gives the total force felt by one wire due to the other as

$$N|\vec{F}| = \frac{(4\pi \times 10^{-7} T \cdot m \cdot A^{-1})(1A)^2(1m)}{2\pi(1m)}$$

$$= \boxed{2 \times 10^{-7} N}$$

since $1 T \cdot A = 1 N$.